

Aim: To write a python code to analyze semantic frames using FrameNet and display frame details such as frame elements and lexical units.

Description: FrameNet is a lexical database based on frame semantics, where: A frame represents a real-world situation , Frame Elements (FEs) are participants in that situation , Lexical Units (LUs) are words that evoke the frame

Program: will first displays random FrameNet frame names, Then asks the user to enter a frame name. Finally shows details of that frame.

```
import nltk
import random
from nltk.corpus import framenet as fn
# Download (run once)
nltk.download('framenet_v17')
frames = list(fn.frames())

print("Total Frames Available:", len(frames))

# Display 10 random frame names
print("\n--- Random Frame Names ---")
random_frames = random.sample(frames, 10)

for f in random_frames:
    print("-", f.name)

# User input
user_input = input("\nEnter a frame name from above: ").strip()

# Display details
try:
    frame = fn.frame(user_input)

    print("\n--- Frame Details ---")
    print("Frame Name:", frame.name)
```

```

print("Definition:", frame.definition)
print("\nFrame Elements:")
for fe in frame.FE:
    print("-", fe)
print("\nLexical Units:")
for lu in frame.lexUnit:
    print("-", lu)
except:
    print("\nInvalid frame name! Please run again and choose from the list.")

```

Output:

otal Frames Available: 1221

--- Random Frame Names ---

- Contingency
- Process_stop
- Visit_host_stay
- Moving_in_place
- Recovery
- Probability
- Motion_scenario
- Attention_getting
- Exemplariness
- Meet_specifications

Enter a frame name from above: Recovery

--- Frame Details ---

Frame Name: Recovery

Definition: These words describe the recovery or healing of a Patient from an Affliction without reference to the influence of any Treatment or Healer (see the Cure frame). ' 'My injuries had not yet completely healed.' 'Henley MP Michael Heseltine, recovering from a heart attack got a very public get well card.'

Frame Elements:

- Patient
- Affliction
- Body_part
- Manner
- Degree
- Means
- Company
- Time
- Place

Lexical Units:

- convalesce.v
- convalescence.n
- heal.v
- recover.v
- recovery.n
- recuperate.v
- recuperation.n
- perk up.v
- ease.v